Junaid Mansur Ifti

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EDUCATION

University Of Dhaka, *Bangladesh* 2018 - 2022
Bachelor of Science in Software Engineering
CGPA: 3.58/4.00 (**3.98/4.00** in Last 2 Years)
Notable Courses: Machine Learning, Artificial Intelligence, Software Requirement Engineering, Software Metrics, Software Security, Software Design and Analysis, Distributed Systems

RESEARCH EXPERIENCE & PUBLICATIONS

Publications under Review

- **Ahmed Ryan, Junaid Mansur Ifti, Md Erfan, Akond Ashfaque Ur Rahman, Md Rayhanur Rahman.** "Between the Lines: A Statement-Level Annotated Dataset and Behavioral Taxonomy for Malicious Python Packages." *Under Review at IEEE Transactions on Software Engineering (TSE)*, 2026. *arXiv:2512.12559*
  Collaboration with researchers at The University of Alabama and Auburn University
  - Built a statement-level annotated dataset of 370 malicious Python packages (833 files, 90,527 lines) with 2,962 labeled occurrences of malicious behavior, and derived a taxonomy of 47 malicious indicators across 7 types.
  - Showed that taxonomy-guided LLM prompting improves malicious package detection precision by 9% on average, reducing false positives.
  - Applied sequential pattern mining to uncover recurring indicator sequences that characterize common attack workflows to strengthen software supply-chain defenses.

Independent Researcher February 2026 – Present
AutoSecuRe: Automated Secret Migration and Benchmarking by LLM | Supervised by Dr. Md Rayhanur Rahman (The University of Alabama) *Ongoing, unpublished*

- Sole researcher: formulated the research problem and designed and implemented an autonomous agentic pipeline that detects hard-coded secrets (Gitleaks with an LLM audit layer to filter mock credentials), migrates them to a secure vault, and refactors source code to use secret-provider calls, going beyond detection-only tools.
- Designed a three-part evaluation framework to benchmark LLMs: functional integrity (unit tests), security completeness (follow-up secret scan), and structural correctness (AST similarity with CodeBERT and Zhang-Shasha against human-verified golden solutions).
- Built and validated an end-to-end proof of concept, including the supporting infrastructure; currently constructing a multi-language benchmark corpus of vulnerable repositories with golden solutions, ahead of large-scale multi-LLM benchmarking.

Undergraduate Researcher, *Distributed Systems & Software Engineering Research Group* 2022
University of Dhaka, Bangladesh

- Developed "GAnomaly", an ML-based web application to identify anomalous web traffic in Google Analytics data, supervised by Dr. Kazi Muheymin.
- Implemented statistical outlier detection techniques and rule-based algorithms based on the IEEE paper "Using Google Analytics to Support Cybersecurity Forensics" to support cybersecurity forensics.

PROFESSIONAL EXPERIENCE

Ding, *Ireland (Hybrid-Remote)*
Senior Software Engineer August 2026 – Present

- Serve as technical lead and team co-supervisor, mentoring junior engineers and accelerating engineering output by integrating AI-assisted coding tools (e.g., Claude Code) into the development workflow.
- Lead critical third-party API integrations, acting as the primary technical liaison between external suppliers and internal C-level directors.
- Spearhead cross-functional initiatives with broader engineering teams to modernize legacy .NET applications, remediate security vulnerabilities, and maintain CI/CD pipeline stability.

Junior Software Engineer May 2023 – August 2026

- Maintained the backend of internal web applications, developed RESTful APIs, and integrated 3rd-party payment APIs (REST/SOAP) into the core web system.

- Led technical meetings and facilitated knowledge transfer sessions for both technical and non-technical stakeholders in a multicultural environment.

SWE Intern, ***Samsung Research & Development Institute, Bangladesh (Onsite)*** January 2022 – June 2022

- Contributed to the core Android codebase (100K+ LOC) using Kotlin and Java, improving and increasing unit test case coverage by 40%.

ACADEMIC & RESEARCH PROJECTS

- GQM Research: Impact of Physical Health on Developer Productivity | *Project Link* 2022
Investigated correlations using the Goal Question Metric approach via manually collected data from software engineers.

- A Tool for Commit Classification | *Project Link* 2022
Developed a Python tool leveraging source code changes and commit messages to categorize maintenance activities into Corrective, Perfective, and Adaptive classes.

- Air Quality Index Predictor (LR) | *Project Link* 2022
Pre-processed scraped custom datasets and applied linear regression to model the correlation between weather patterns and AQI in Chittagong.

- Control Flow Graph and Software Metrics Tool | *Project Link* 2022
Built a Python tool to compute Halstead complexity and LOC for automated software quality assessment.

- Stock Price Prediction (LR) | *Project Link* 2022
Developed a linear regression model to predict stock market trends using historical data within the context of Bangladesh.

- AI Game Agents (Connect 4 & Wumpus World) 2021
Designed inferential logic agents and applied the Minimax algorithm with Alpha-Beta pruning for heuristic evaluation and decision-making.

- Machine Learning Classifiers From Scratch 2021
Built a K-Nearest Neighbors (KNN) algorithm to predict divorce likelihood based on psychological datasets, and a Naive Bayes classifier for lightweight skin pixel detection in images.

SKILLS

- **Programming:** Python, C#, C++, Java, JavaScript, PHP, Kotlin
- **Frameworks & Technologies:** .NET Core, Flask, pandas, scikit-learn, matplotlib, MongoDB, MSSQL, NUnit, JUnit
- **Tools & Platforms:** Jenkins, Git, Docker, Docker-Compose, Jupyter Notebook, Nginx
- **Problem Solving Profile**: **(Leetcode) bsse1027 | Solved Problems: 111 | Max Contest Rating: 1463**

STANDARDIZED TEST SCORES

IELTS Academic: Listening 8.5 | Speaking 8 | Writing 7.5 | Reading 8.5 | **Overall 8**

ACCOMPLISHMENTS & INVOLVEMENT

- **National Cyber Drill by BGD e-Gov CIRT:** Secured a spot in the Top 20 among 234 teams in a national CTF competition (2020).
- **Awards:** Government Scholarship for Higher Secondary (2017) and Secondary Certificate Exam (2015); Bangladesh Physics Olympiad Divisional Medalist (2012).
- **IIT Software Engineers' Community (2019-2022):** Served as Organizing & Cultural Secretary; led technical teams for "ITVerse 1.0" and operational teams for "TechCrunch 2.0," an ADB-funded workshop promoting women in IT.
- **Fight For Life Initiative (2020):** Volunteered to create an integrated payment website to raise emergency funds for poverty-stricken individuals during the initial Covid-19 lockdown.

REFERENCES

Dr. Md Rayhanur Rahman
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